PromptRad: Knowledge-Enhanced Multi-Label Prompt-Tuning for Low-Resource Radiology Report Labeling

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Research Background

- Early-stage diagnosis for liver cancer is challenging due to its complexity and diversity of tumor features [1].
- Benign features can act as indicators for early-stage tumors.
- In some hospitals (e.g., CGMH), only malignant features are directly retrievable from the cancer registry.

[1] Zhen, S. H., Cheng, M., Tao, Y. B., Wang, Y. F., Juengpanich, S., Jiang, Z. Y., Jiang, Y. K., Yan, Y. Y., Lu, W., Lue, J. M., Qian, J. H., Wu, Z. Y., Sun, J. H., Lin, H., & Cai, X. J. (2020). Deep Learning for Accurate Diagnosis of Liver Tumor Based on Magnetic Resonance Imaging and Clinical Data. Frontiers in oncology, 10, 680. https://doi.org/10.3389/fonc.2020.00680

Radiology report labeling task

Multi-label Classification

Input: radiology report

A 75 Y/O male; Clinical Information:LC-C, Increasing AFP, 2.9> 6.4 > 17.0, RFA on 105-8-2, Please evaluate any recurrenceCT scan of liver for F/U a patient of HCC post RFA was done by using triphasic study without and with bolus IV non-ionic contrast enhancement showed: ** Comparison CT study: 2016-06-24; 2017-3-23 ** 1. Mildly undulated surface of liver with hypertrophy of lateral segment. No ascites. Mild splenogemaly with EVs. Cirrhosis of liver with portal hypertension is considered. 2. Hypodense lesion about 4.4x3.5-cm with lobulated margin in S7, C/W post RFA change. High suspicion of newly developed enhancing viable tumor at posterior aspect with arterial wash-in (se 7, im 15-16). 3. No new liver tumor is found. A small cyst in S4.4. The portal and hepatic venous system are patent. No biliary tree dilatation. 5. No remarkable finding in the gallbladder, pancreas and both kidneys, adrenal glands. 6. No evidence of enlarged lymph node is found in the perigastric area, hepatoduondenal ligament, para-aortic area, pelvis and inguina. 7. Grossly, no abnormality is found in the GI tract. Clear mesentery and omentum. No ascites. 8. Normal contour, capacity and wall thickness of urinary bladder. Normal size and contour of seminal vesicle and prostatic gland. 9. Clear retroperitoneum. Normal contour, diameter and enhancement of abdominal aorta. 10. No active lesion is found in the visualized bilateral lower lungs. 11. Grossly, no destructive bony lesion or abnormal bone density. IMP: Cirrhosis of liver with portal hypertension is noted. A HCC in S7 post RFA with high suspicion of small viability at posterior aspect. No new HCC is found.

Output: categories in interest

- Cyst ✓
- HCC ✓
- Post-treatment ✓
- Cirrhosis ✓
- Steatosis
- Metastasis
- Hemangioma

Related work

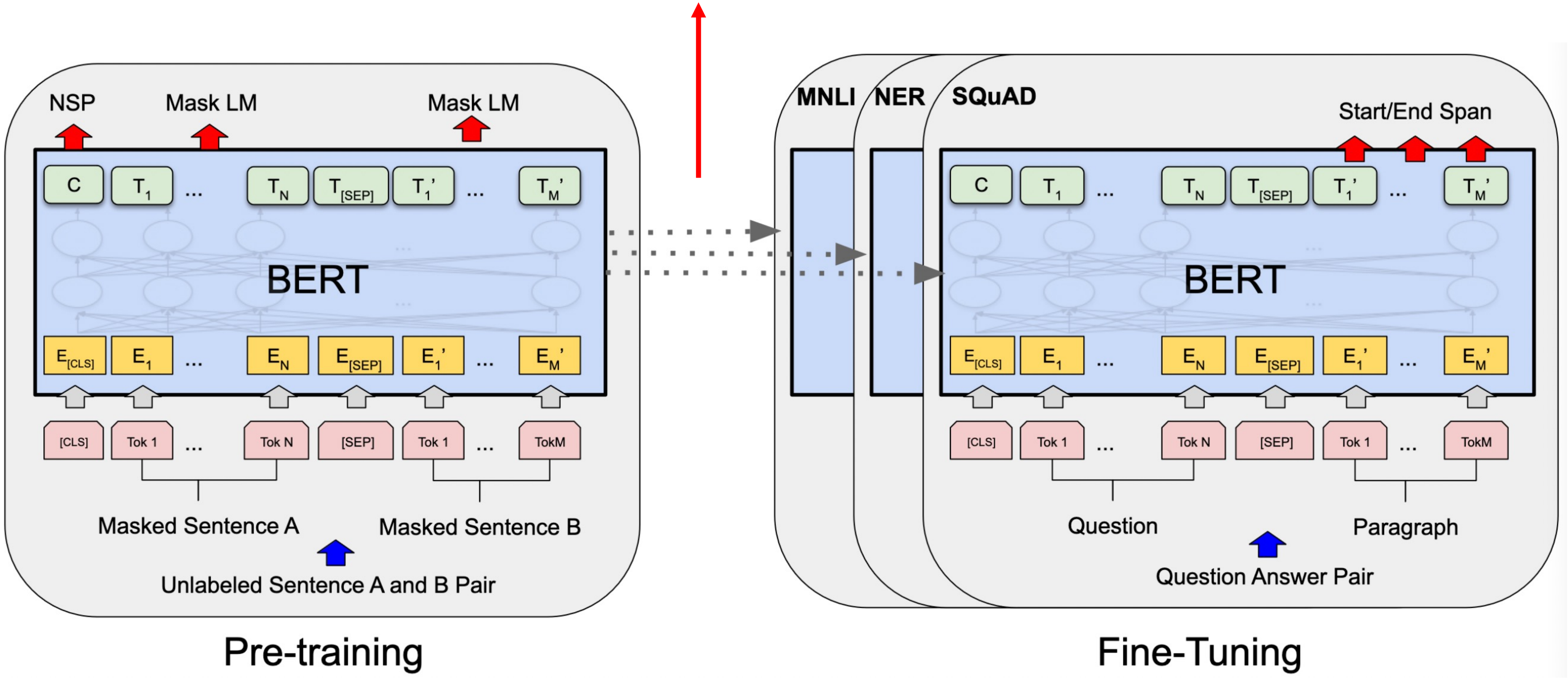
- Free-text reports cannot be easily parsed by pre-defined rules.

Dictionary-based Methods	Such as MetaMap [2], NegBio [3]	**Fail to cover a wide variety of medical descriptions**
Machine Learning Methods	Such as BioBERT [4], PubMedBERT [5]	**Require much human annotated data for training**

[2] Aronson, Alan R. "Effective mapping of biomedical text to the UMLS Metathesaurus: the MetaMap program." Proceedings of the AMIA Symposium. American Medical Informatics Association, 2001.
[3] Peng, Yifan et al. “NegBio: a high-performance tool for negation and uncertainty detection in radiology reports.” AMIA Joint Summits on Translational Science proceedings. AMIA Joint Summits on Translational Science vol. 2017 188-196. 18 May. 2018
[4] Lee, Jinhyuk, et al. "BioBERT: a pre-trained biomedical language representation model for biomedical text mining.” Bioinformatics 36.4 (2020): 1234-1240.
[5] Gu, Yu, et al. "Domain-specific language model pretraining for biomedical natural language processing." ACM Transactions on Computing for Healthcare (HEALTH) 3.1 (2021): 1-23.

BERT [6] requires much data for fine-tuning

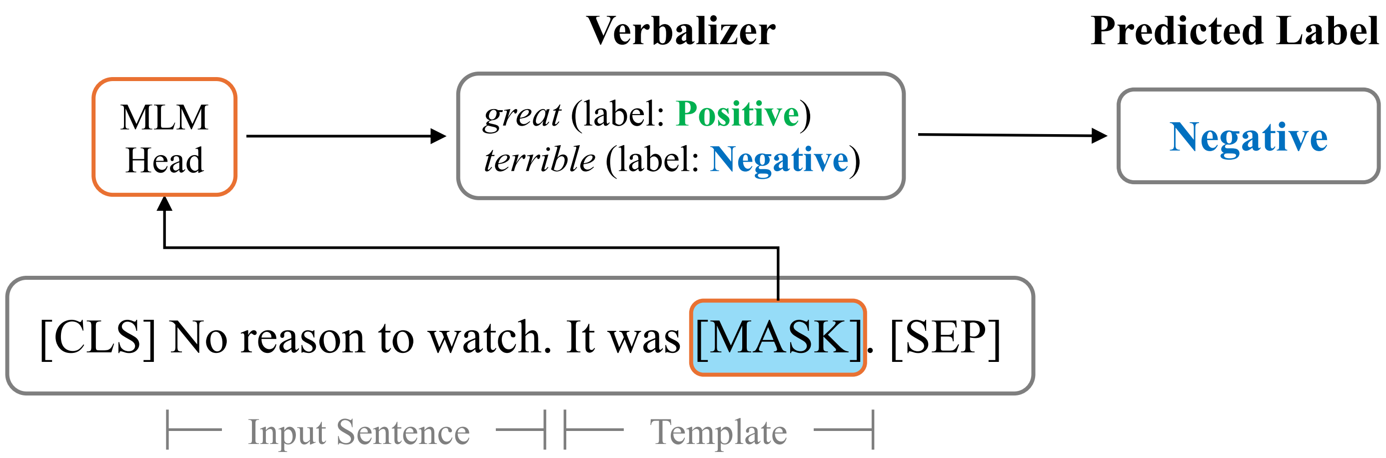
Additional Classification head (layer) ➡ requires much labeled data[7][8]



[6] Devlin, Jacob, et al. "Bert: Pre-training of deep bidirectional transformers for language understanding." NAACL (2019).
[7] Dodge, Jesse, et al. "Fine-tuning pretrained language models: Weight initializations, data orders, and early stopping." arXiv preprint arXiv:2002.06305 (2020).
[8] Tianyi Zhang, , Felix Wu, Arzoo Katiyar, Kilian Q Weinberger, Yoav Artzi. "Revisiting Few-sample BERT Fine-tuning." ICLR (2020).

Prompt-Tuning (Prompt-based Fine-Tuning)

- Schick and Schütze (2021)[9], Gao et al. (2021)[10] proposed to adopt the pre-trained masked language model (MLM) for downstream tasks.



➡ **Cannot be directly applied to multi-label classification**

[9] Schick, Timo, and Hinrich Schütze. "Exploiting cloze questions for few shot text classification and natural language inference."EACL (2021).
[10] Gao, Tianyu, Adam Fisch, and Danqi Chen. "Making Pre-trained Language Models Better Few-shot Learners." ACL (2021).

Proposed method: Multi-label Prompt-Tuning for Radiology Report Labeling (PromptRad)

Problem Definition

- Given $D_{train} = \{(r_a, \mathcal{Y}_a)\}_{a=1}^{K}$, with the label space $\mathcal{Y} = \{y_1, y_2, \dots y_n\}$, our task is to label radiology reports with findings $y_1, y_2, \dots y_n$, under the scenario where only K labeled reports are available for training.
- r_a is the a-th report in D_{train}, and $\mathcal{Y}_a \subseteq \mathcal{Y}$ is the set of positive findings in r_a.

	y_1	y_2	y_3	y_4	y_5	y_6	y_7
Liver Cancer-related Categories	Cyst	Hepato-cellular Carcinoma	Post-treatment	Cirrhosis	Steatosis	Metastasis	Hemangioma

Multi-word Verbalizer

- Category to label words (Closest word + Synonyms from SNOMED CT of UMLS)
  - UMLS: **U**nified **M**edical **L**anguage **S**ystem
  - SNOMED CT: **S**ystematized **N**omenclature of **Med**icine -- **C**linical **T**erms

Closest word + Synonym

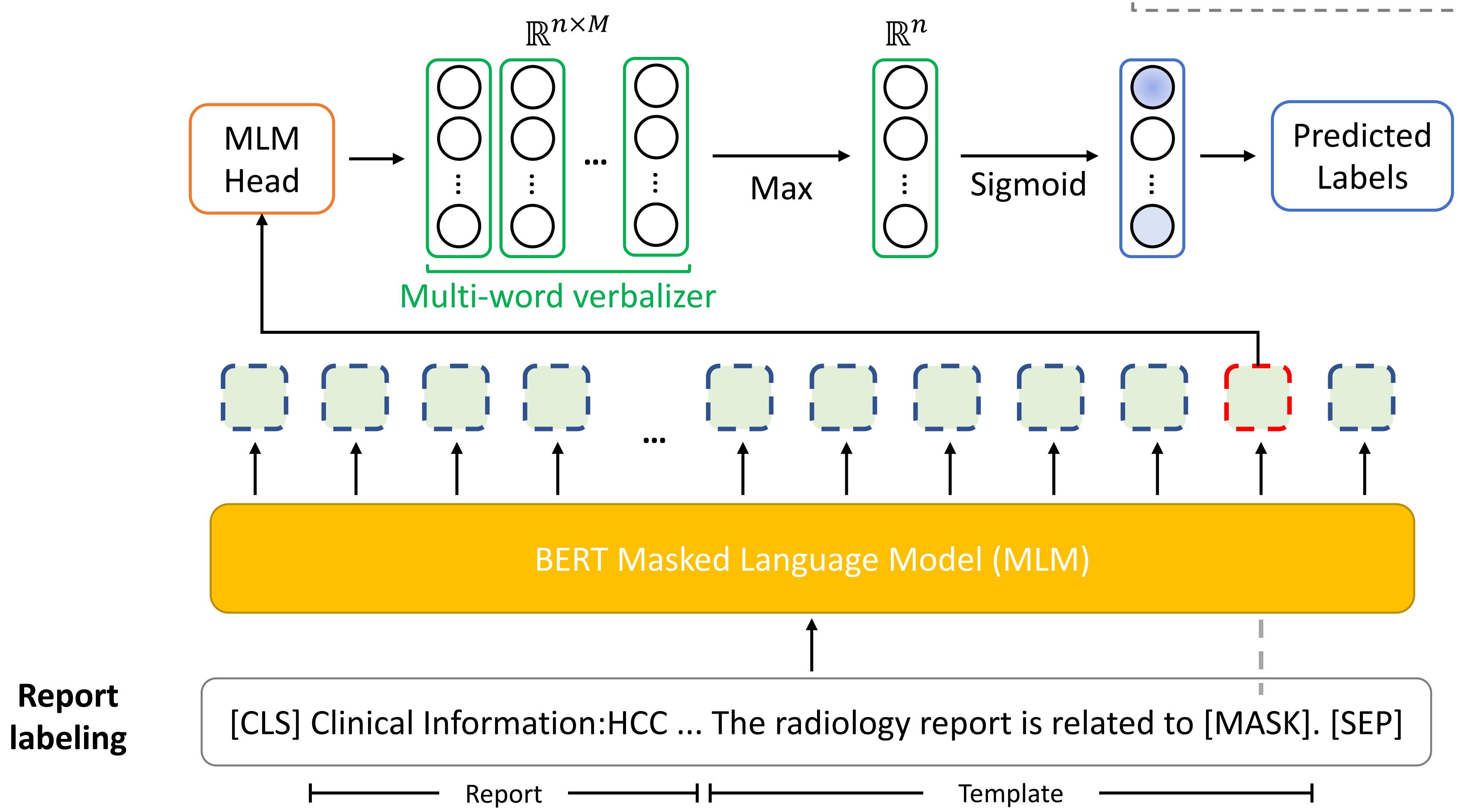
cyst (label: Cyst)
hcc, hepatoma (label: Hepatocellular Carcinoma)
steatosis, steatohepatitis (label: Steatosis)
...

Report labeling

[CLS] Clinical Information:HCC ... The radiology report is related to [MASK]. [SEP]

Workflow of PromptRad

n: number of labels
M: maximal number of mappings among all categories



Automatic Prompt Generation with T5 [10,11]

[10] Gao, Tianyu, Adam Fisch, and Danqi Chen. "Making Pre-trained Language Models Better Few-shot Learners." ACL (2021).
[11] Raffel, Colin, et al. "Exploring the limits of transfer learning with a unified text-to-text transformer." The Journal of Machine Learning Research 21.1 (2020): 5485-5551.

Original sentence

Thank you for inviting me to your party last week.

Corrupted sentence

Thank you <x> me to your party <y> week. → T5 → <x> for inviting <y> last <z>

[CLS] Clinical Information:HCC ... The radiology report is related to [MASK]. [SEP]

Report | Template

Format 1 (**Input report** first): **[Report]** <x> [ANS] <y>

Format 2 (**Placeholder** first): <x> **[ANS]** <y> [Report]

Experiments

Dataset

- We use de-identified **liver** computed tomography (**CT**) radiology **reports**, which are from the Chang Gung Memorial Hospital (CGMH), Linkou, Taiwan.
- The reports were annotated by two senior radiologists from CGMH (co-authors).

Seven common categories related to **liver** cancer

Categories	Training Set (2008-2014)	Test Set (2015-2017)
Cyst	275	109
Hepatocellular Carcinoma	214	79
Post-Treatment	205	79
Cirrhosis	168	85
Steatosis	154	93
Metastasis	101	46
Hemangioma	90	29
Total number of reports	773	325

Performance comparison in F1-score using the test set

Table 2: Performance comparison in F1-score (%) using the independent test set.

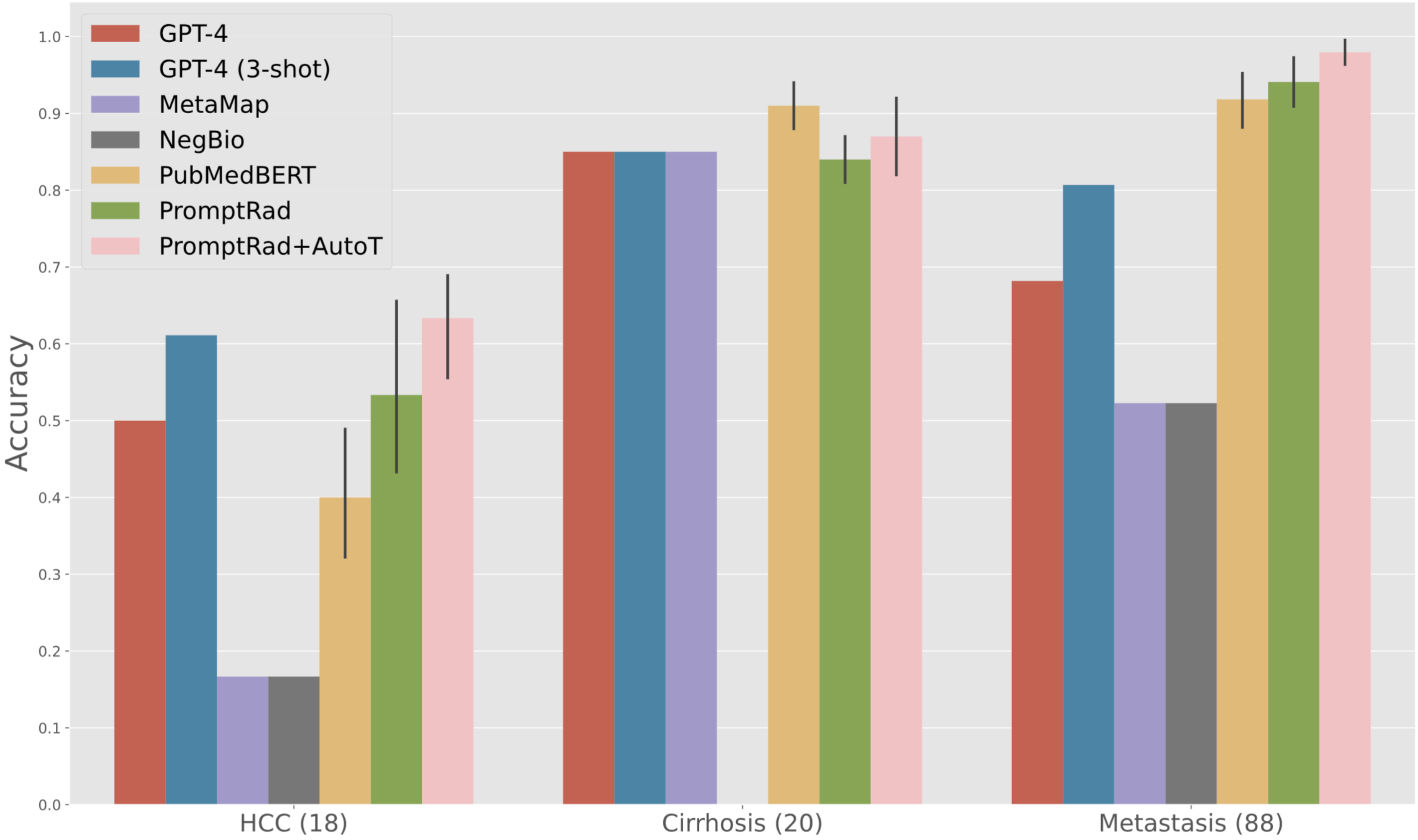
	Cyst	HCC	Post-T	Cirr.	Stea.	Meta.	Hem.	Mac. F1	Mic. F1
GPT-4	86.1	91.5	79.1	96.6	98.9	73.8	95.1	88.7	88.7
GPT-4 (ICL[a])	89.7	88.8	76.3	92.5	92.0	78.5	94.9	87.5	87.4
Label Match	67.5	88.5	0.0	87.7	0.0	48.6	**98.3**	55.8	62.3
MetaMap (MM)	77.6	86.9	48.6	53.5	95.7	27.5	84.6	67.8	69.1
NegBio (NB)	77.6	86.9	**81.4**	82.7	95.7	27.5	84.6	76.6	79.2
PMB[b]	$53.7_{8.4}$	$80.4_{5.2}$	$70.4_{4.3}$	$64.5_{12.7}$	$48.3_{12.9}$	$54.9_{18.1}$	$37.7_{21.7}$	$58.6_{10.0}$	$60.9_{8.2}$
PMB[b]+MM	$68.1_{5.4}$	$83.9_{1.7}$	$64.5_{2.3}$	$68.0_{6.6}$	$84.5_{6.7}$	$47.1_{10.0}$	$70.1_{15.2}$	$69.5_{5.0}$	$70.3_{3.8}$
PMB[b]+NB	$68.1_{5.4}$	$83.9_{1.7}$	$76.8_{0.3}$	$81.0_{3.7}$	$84.5_{6.7}$	$47.1_{10.0}$	$70.1_{15.2}$	$73.1_{4.5}$	$74.1_{3.5}$
PR[b]	$78.0_{5.3}$	$89.1_{0.8}$	$76.9_{1.8}$	$86.2_{4.2}$	$95.7_{3.7}$	$71.9_{3.5}$	$88.4_{5.6}$	$83.7_{2.1}$	$84.1_{1.7}$
PR+AutoT[b]	$\mathbf{89.5}_{2.4}$	$\mathbf{90.8}_{1.0}$	$78.4_{3.9}$	$\mathbf{91.0}_{3.4}$	$\mathbf{97.3}_{0.9}$	$\mathbf{84.7}_{1.2}$	$92.4_{2.5}$	$\mathbf{89.2}_{1.0}$	$\mathbf{89.4}_{1.0}$

Train_size=32

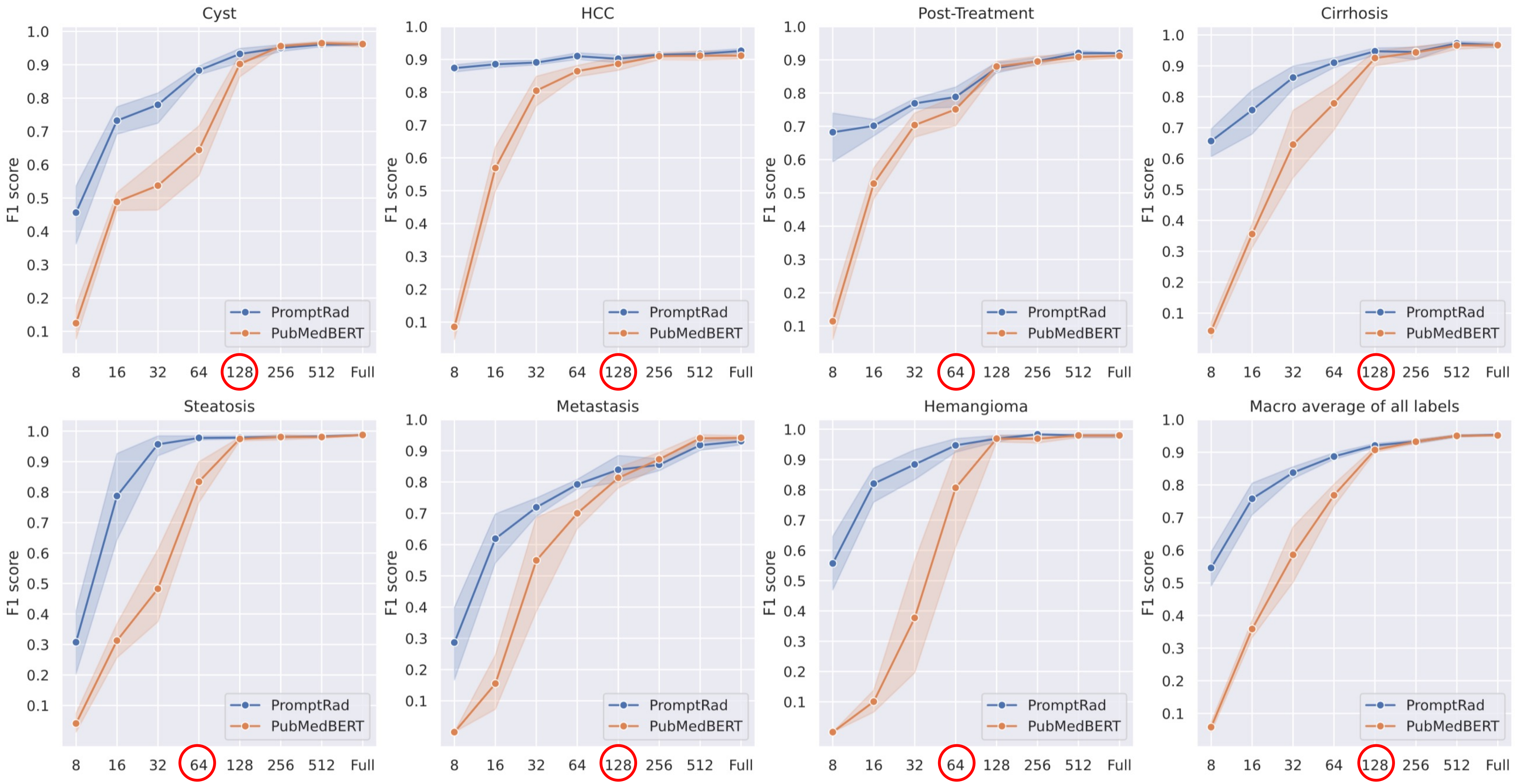
Abbreviations: Cirr.: Cirrhosis; Stea.: Steatosis; Meta.: Metastasis; Hem.: Hemangioma; Mac.: Macro average; Mic.: Micro average; Post-T: Post-Treatment; PMB: PubMedBERT; MM: MetaMap; NB: NegBio; PR: **P**rompt**R**ad (ours).
[a] In-Context Learning with three random samples from the training set. [b] Average from five runs; subscripts: standard deviation.

Quantitative Results for the Negation Experiment

- Diverse negative descriptions
  - No ... nor HCC in both lobes liver.
  - R/O metastasis
- Positive mentions but negative labels.



Using More Training Data?



Summary of PromptRad

- PromptRad outperforms the other baselines for report labeling, except GPT-4.
- Compared with GPT-4, the proposed method with AutoT is competitive and suitable for data sensitive report labeling.
- Compared with dictionary-based methods, PromptRad is better handling negation cases.

Thank you!

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github.com/ila-lab/PromptRad